

ENTERPRISE DATA PLATFORM · TRAINING SERIES

Big Data & Databricks

From Fundamentals to the Lakehouse

A consulting-grade journey through modern data architecture — why Big Data matters, what the Databricks Lakehouse is, and how it is engineered end to end.

01 Big Data Fundamentals

02 Introduction to Databricks

03 Databricks Architecture

Audience: Data Engineers · Analysts · Data Scientists · Cloud Architects · Technical Leaders

Powered by the Databricks Lakehouse Platform · Apache Spark · Delta Lake · Unity Catalog

Your Learning Journey



SECTION 01

Big Data Fundamentals

- Why Big Data matters
- The 5 V's & challenges
- Technologies & ecosystem
- Real-world use cases

OUTCOME *Speak the language of Big Data with confidence.*



SECTION 02

Introduction to Databricks

- What & why Databricks
- The Lakehouse concept
- Unified analytics platform
- Core components & benefits

OUTCOME *Explain the Lakehouse and what Databricks unifies.*



SECTION 03

Databricks Architecture

- Control & compute planes
- Spark, Delta & Medallion
- Clusters, security, data flow
- Integration architecture

OUTCOME *Describe how Databricks runs end to end.*

Twelve concepts · three sections · one continuous narrative — from raw data to a governed, AI-ready Lakehouse.

SECTION ONE

Big Data Fundamentals

- Why Big Data matters & the 5 V's
- Traditional vs Big Data & key challenges
- Technologies, ecosystem & real-world use cases



What Is Big Data?

DEFINITION

Big Data refers to datasets so large, fast-moving, and varied that traditional databases and single-server tools can no longer capture, store, or process them — requiring distributed, horizontally-scalable systems to turn raw signals into value.

- Not just size — it is volume, speed, and diversity combined
- Powered by distributed computing across clusters of machines
- 80%+ of new data is unstructured: text, images, logs, video, IoT

402M+

terabytes of data
created every day

181 ZB

global data volume
by 2025

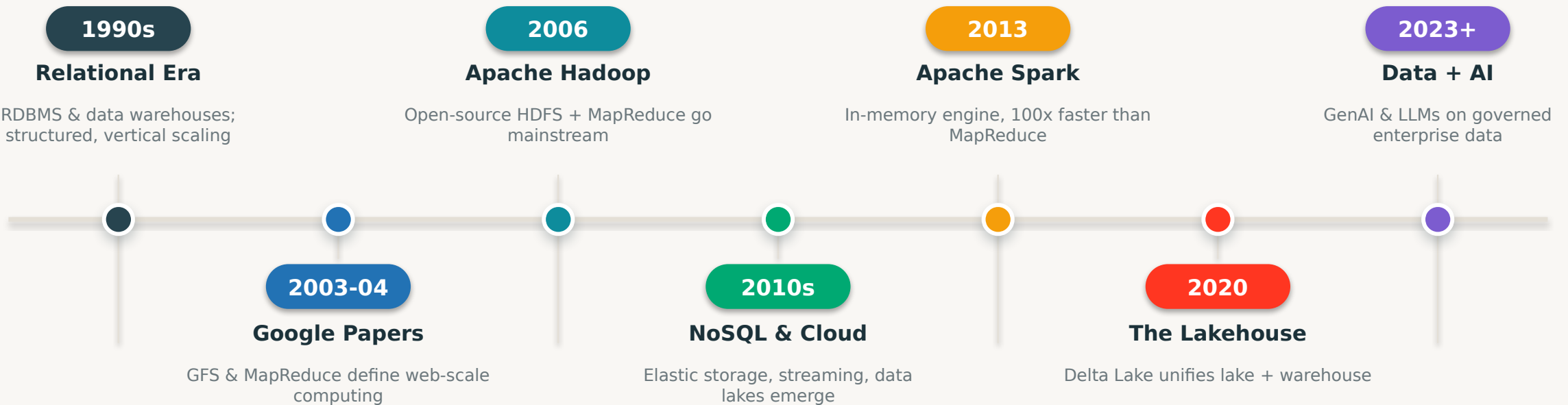
80%+

of enterprise data is
unstructured

5G · IoT

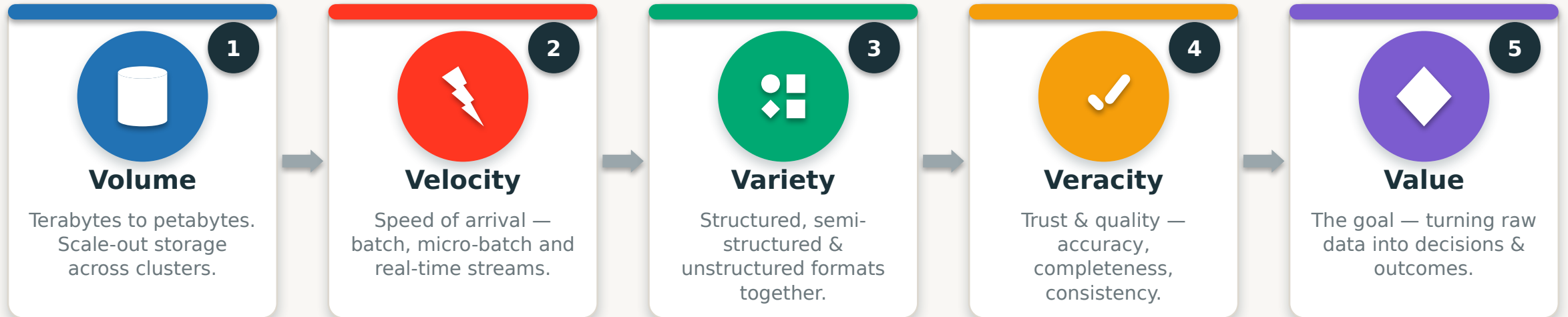
billions of connected
sensors & devices

The Evolution of Big Data



Each wave solved the prior era's bottleneck — scale, then speed, then unification, and now intelligence.

The 5 V's of Big Data



Volume · Velocity · Variety describe the data; Veracity · Value describe whether it can be trusted and what it returns.

Enterprise lens **Volume:** years of transactions **Velocity:** card swipes per second **Variety:** JSON, images, clickstream **Veracity:** duplicate / missing fields **Value:** real-time fraud blocked

Traditional Data vs. Big Data

Aspect	Traditional Data	Big Data
Volume	Gigabytes to terabytes	Terabytes to petabytes & beyond
Data Types	Structured rows & columns	Structured + semi + unstructured
Schema	Schema-on-write (rigid)	Schema-on-read (flexible)
Processing	Centralized, single server	Distributed across clusters
Scaling	Scale up — bigger machine	Scale out — more machines
Architecture	Monolithic RDBMS / warehouse	Data lake / Lakehouse, decoupled
Cost Model	Costly proprietary hardware	Commodity & elastic cloud

Big Data Challenges



Storage & Scale

Cost-effectively storing petabytes across distributed, fault-tolerant systems.



Data Quality

Duplicates, missing values and inconsistency erode trust (the Veracity problem).



Integration & Silos

Hundreds of sources and formats fragmented across disconnected systems.



Real-Time Processing

Acting on streaming events in milliseconds, not overnight batch windows.



Skills & Complexity

Scarce talent and brittle, hand-stitched toolchains slow delivery.



Governance & Security

Privacy, access control, lineage and regulatory compliance (GDPR, HIPAA).

The Big Data Technology Landscape



Ingestion

Kafka · Spark Structured Streaming ·
Fivetran · NiFi · Auto Loader



Storage

HDFS · Amazon S3 · ADLS · Google Cloud
Storage · Delta Lake



Processing

Apache Spark · Flink · MapReduce · Photon
· dbt



Orchestration

Apache Airflow · Databricks Workflows ·
Dagster



Query & Analytics

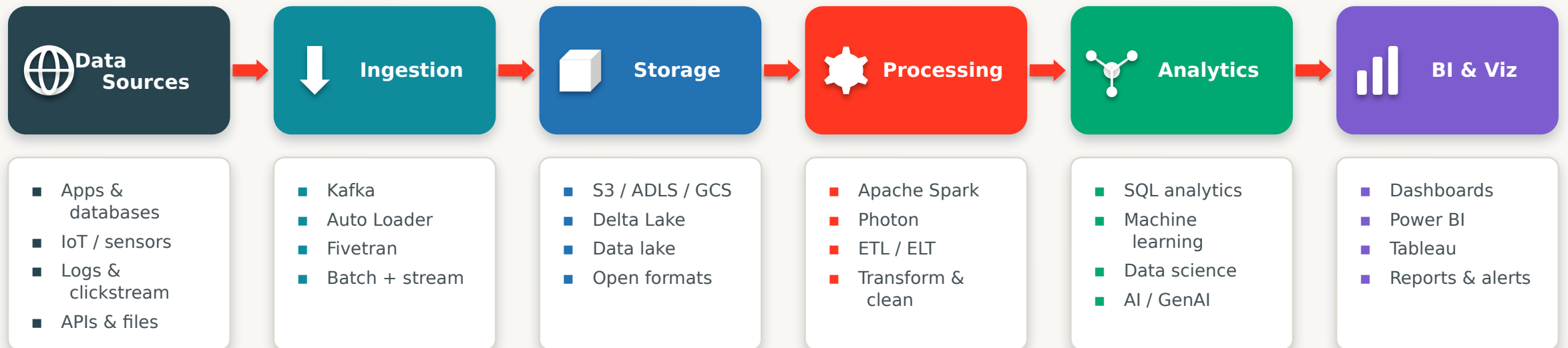
Hive · Trino / Presto · Databricks SQL ·
Snowflake



ML & AI

MLflow · TensorFlow · PyTorch · scikit-learn ·
Mosaic AI

The Big Data Ecosystem — End to End



*Raw signals flow left-to-right, gaining structure and value at each stage — the same backbone the Databricks Lakehouse implements as **one unified platform.***

Big Data in the Real World



Financial Services

Real-time fraud detection, risk modelling & algorithmic trading on streaming transactions.



Healthcare & Life Sciences

Genomics at scale, medical imaging AI and predictive patient diagnostics.



Retail & E-Commerce

Personalized recommendations, demand forecasting & dynamic pricing.



Manufacturing

Predictive maintenance and IoT sensor analytics across the factory floor.



Telecom & Media

Network optimization, churn prediction and content recommendation.



Energy & Mobility

Smart-grid balancing, fleet telematics and autonomous-vehicle data.

SECTION TWO

Introduction to Databricks

- What & why Databricks · the evolution
- The Lakehouse concept & unified platform
- Core components & enterprise benefits

What Is Databricks?

DEFINITION

Databricks is a unified **Lakehouse platform** that combines data engineering, data warehousing, data science, analytics, AI and machine learning in a single collaborative environment — built on open standards and powered by **Apache Spark and Delta Lake**.

WHY THE WORDING MATTERS

Vague: *“Databricks is a cloud platform used for analytics.”*

Precise: *“A unified Lakehouse uniting engineering, warehousing, science, AI & ML on one collaborative, Spark-powered platform.”*



Born from Spark

Founded 2013 by the creators of Apache Spark.



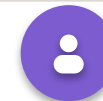
Multi-Cloud

Runs natively on AWS, Azure and Google Cloud.



Open-Source Roots

Created Spark, Delta Lake, MLflow & Unity Catalog.



One Platform

Every data persona in one governed workspace.

Why Databricks? The Problem It Solves

THE TRADITIONAL TWO-SYSTEM WORLD

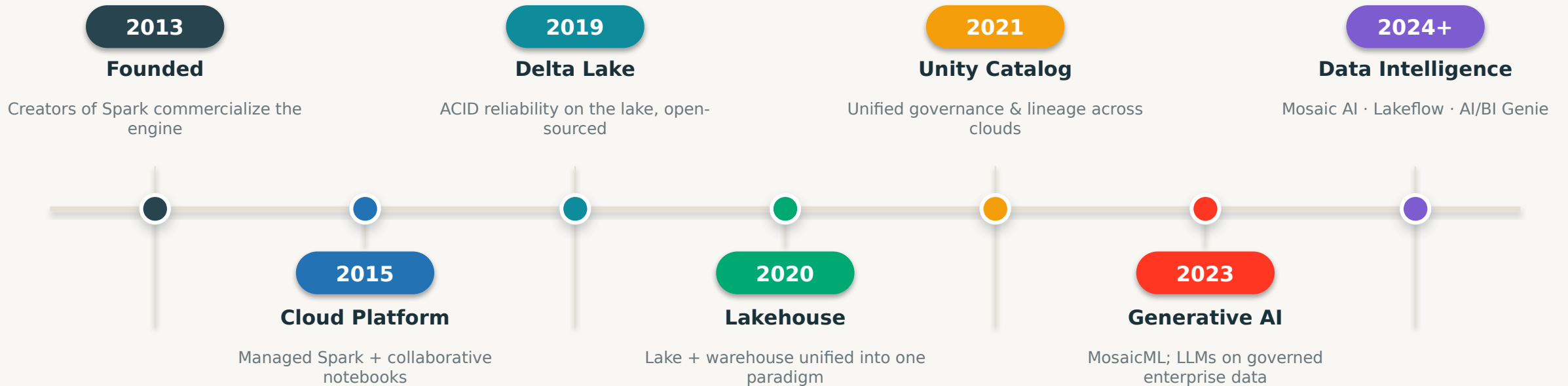
- Separate data lake AND data warehouse to maintain
- Constant, brittle copying of data between systems
- Duplicated storage, stale copies, drifting definitions
- Governance & security bolted on per system
- ML on the lake, BI on the warehouse — never together
- High cost, high complexity, slow time-to-insight



THE DATABRICKS LAKEHOUSE ANSWER

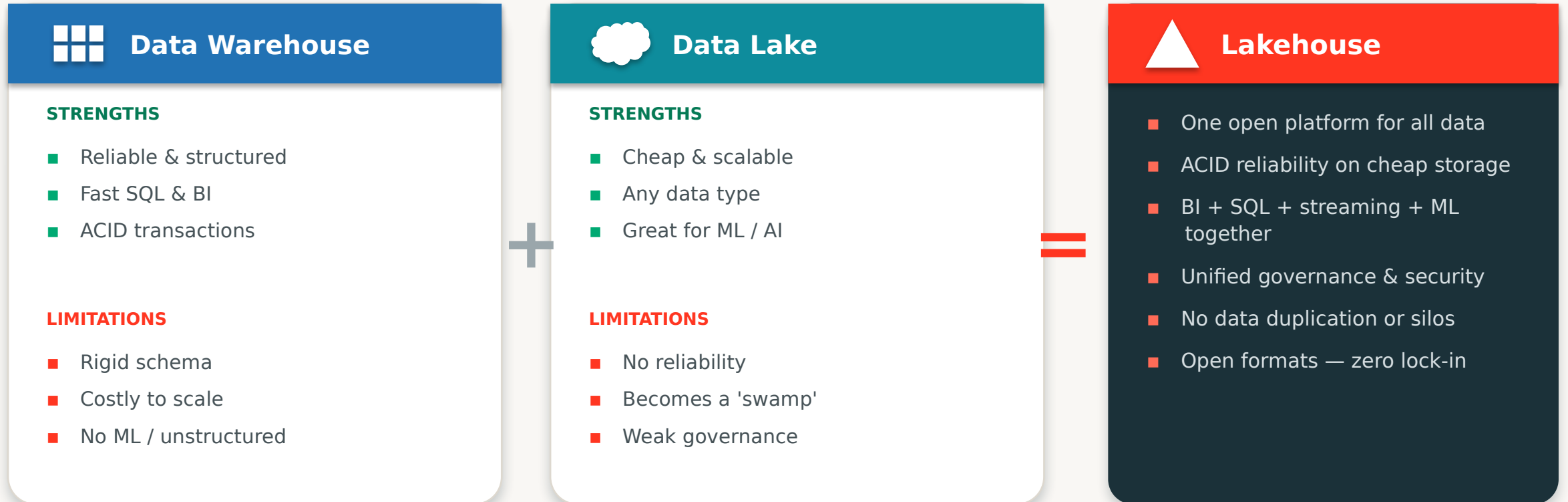
- One platform for ALL data, analytics & AI
- Open Delta Lake storage — single copy, no silos
- ACID reliability directly on cheap cloud storage
- Unity Catalog: one governance & security model
- BI, SQL, streaming & ML on the same data
- Lower cost, less complexity, faster insight

The Databricks Evolution



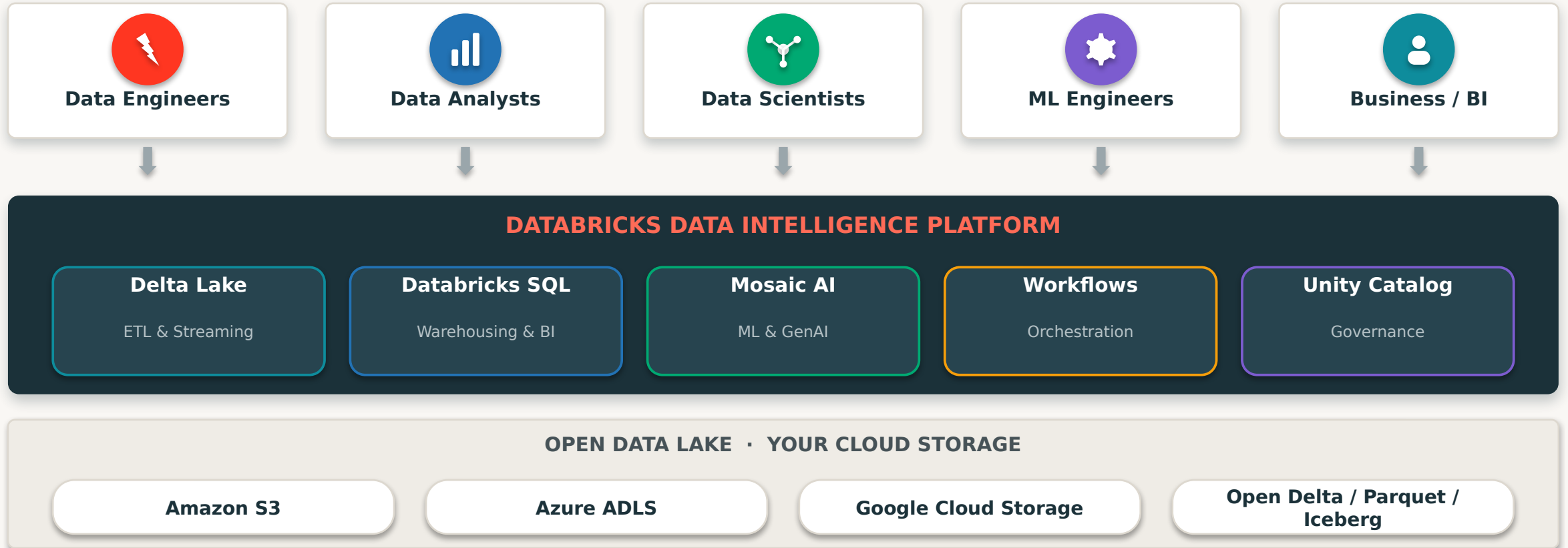
From a faster engine, to reliable storage, to a unified platform, to a Data Intelligence Platform for the AI era.

The Lakehouse — Best of Both Worlds



Lakehouse = the reliability & performance of a warehouse + the scale, openness & flexibility of a lake

One Unified Platform, Every Persona



Databricks Core Components



Together these form the Lakehouse: the Workspace is where people work, Spark + Photon is the engine, Delta Lake is the reliable storage, Mosaic AI delivers intelligence, and Unity Catalog governs it all — one secure, open, collaborative system.

Why Enterprises Choose Databricks



Unified

Engineering, analytics, ML & AI on one platform — no tool sprawl.



Open & Standard

Delta, Spark, MLflow, Iceberg — open formats, zero vendor lock-in.



High Performance

Photon engine & optimized runtime deliver record price/performance.



Elastic Scale

Serverless & autoscaling compute from gigabytes to petabytes.



Collaborative

Shared notebooks, Git integration and reproducible workflows.



Governed & Secure

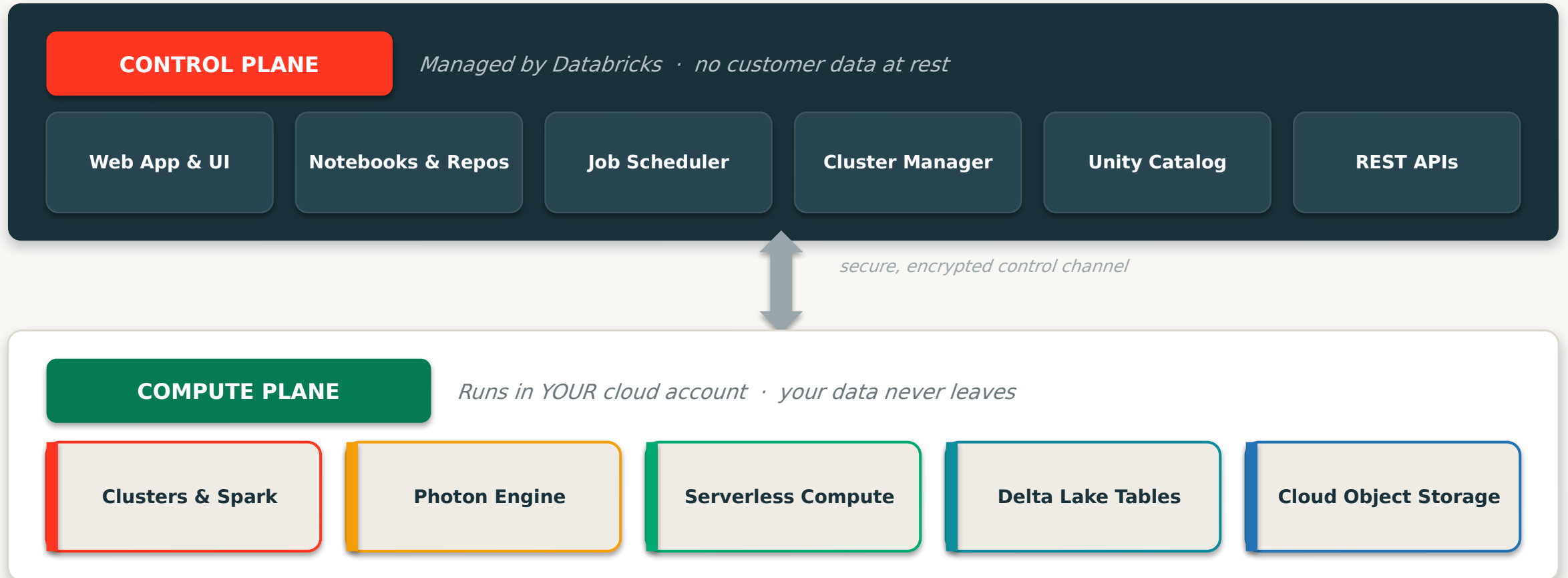
Unity Catalog, fine-grained access, lineage and compliance built in.

SECTION THREE

Databricks Architecture

- Control & compute planes · Spark internals
- Delta Lake, medallion & cluster architecture
- Security, end-to-end data flow & integration

Architecture Overview — Two Planes



Control Plane & Compute Plane in Detail

CONTROL PLANE · Databricks-Managed

Hosts the experience & metadata

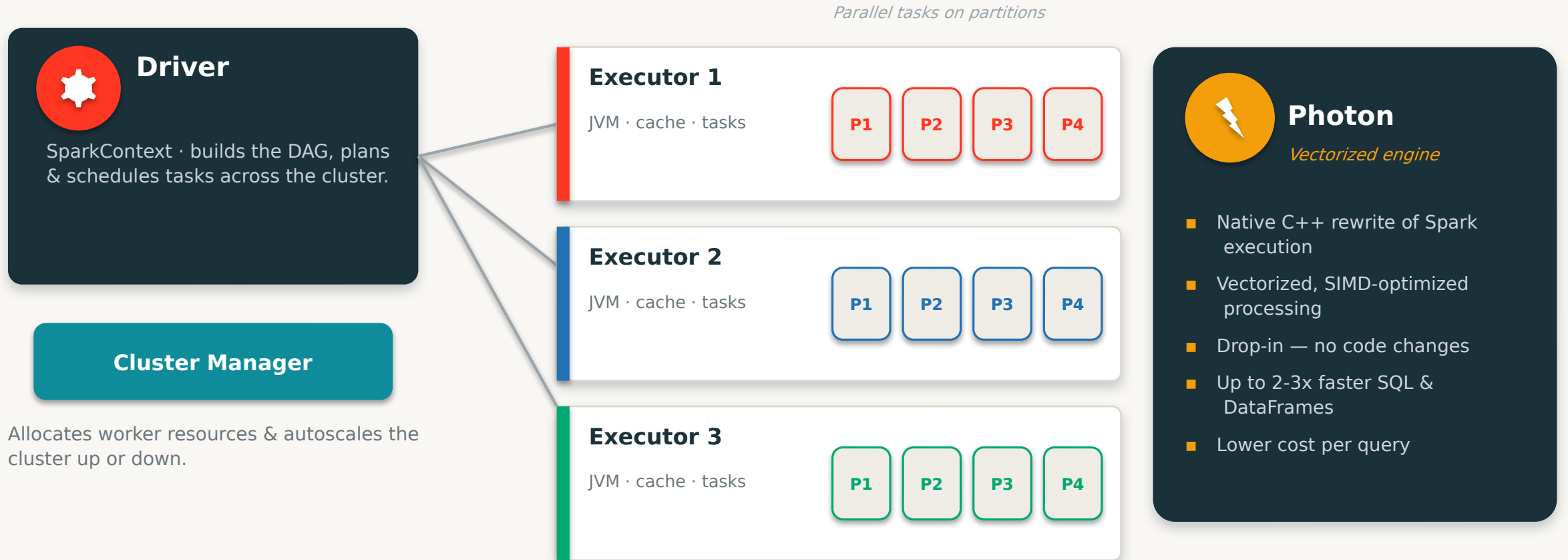
- Web application, UI & collaborative notebooks
- Jobs scheduler & workflow orchestration
- Cluster lifecycle manager & autoscaling logic
- Unity Catalog metastore — permissions & lineage
- REST APIs, Git integration & connectors
- No customer data stored at rest

COMPUTE PLANE · Your Cloud Account

Where data is actually processed

- Clusters of VMs running Apache Spark + Photon
- Reads & writes Delta tables in your object storage
- Classic compute — in your own cloud VPC/VNet
- Serverless compute — instant, Databricks-hosted pool
- Your data & credentials never leave your tenancy
- Encrypted, network-isolated, compliance-ready

Apache Spark — Distributed Execution



Delta Lake — Reliability on the Lake

HOW IT WORKS

Cloud Object Storage



_delta_log/ transaction log → ACID, versions, schema

= One reliable, governed Delta TABLE

Parquet data files + a JSON transaction log turn a folder of files into a database-grade table.



ACID Transactions

Reliable concurrent reads & writes — no corrupt, partial data.



Time Travel

Query or roll back to any previous version of the table.



Schema Enforcement

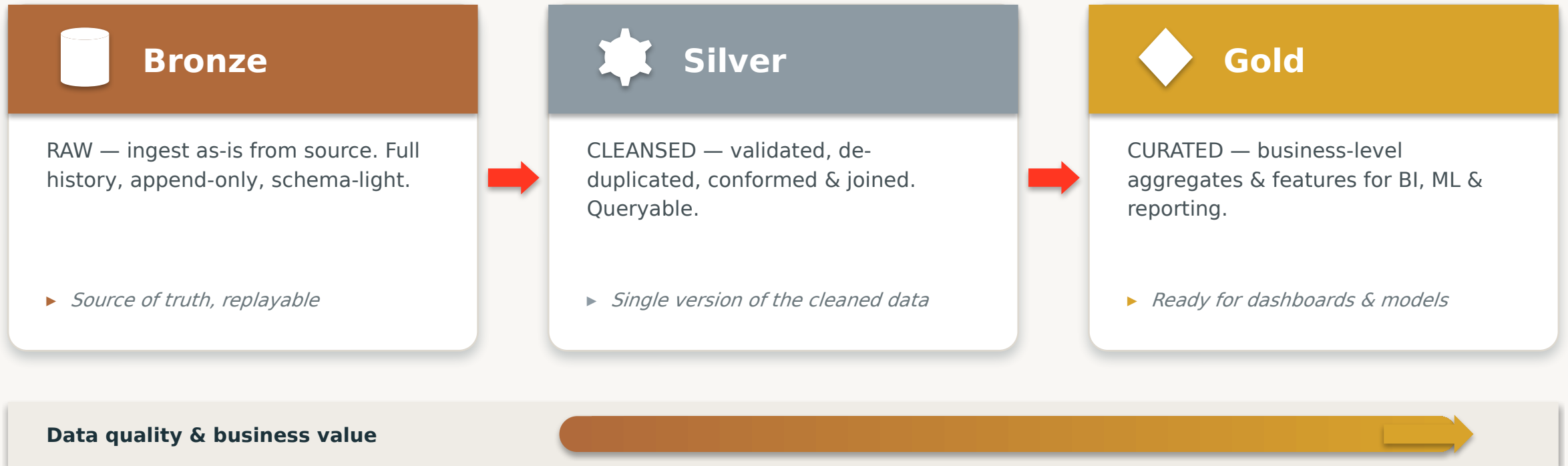
Rejects bad data; supports safe schema evolution.



Batch + Streaming

One table for both, with MERGE/upsert & OPTIMIZE / Z-ORDER.

The Medallion Architecture



A proven, incremental refinement pattern — raw data is progressively cleaned and enriched into trusted, analytics-ready tables.

Cluster & Compute Architecture

ANATOMY OF A CLUSTER

Driver Node · coordinates the cluster

Worker 1

Worker 2

Worker 3

+ auto

Autoscaling *adds/removes workers with load*

Databricks Runtime (DBR)

- Pre-tuned Spark + Photon + Delta + libraries
- ML runtime variant bundles PyTorch / TF / CUDA



All-Purpose

Interactive, shared clusters for collaborative development in notebooks.



Job Clusters

Ephemeral compute created for a job and torn down after — cost-efficient.



SQL Warehouses

Optimized Photon compute for BI & SQL analytics workloads.



Serverless

Instant, fully-managed pools — no infrastructure to configure.

Security & Governance — Unity Catalog

ONE NAMESPACE, ALL ASSETS

Catalog *top-level container*

Schema *database / namespace*

Table / View / Volume / Model *the governed asset*

catalog . schema . table

One consistent name & permission across all clouds & workspaces



Fine-Grained Access

Row/column-level security, ANSI SQL GRANTS, identity & SSO integration.



Automated Lineage

Column-level lineage and audit logs across notebooks, jobs & dashboards.



Discovery & Sharing

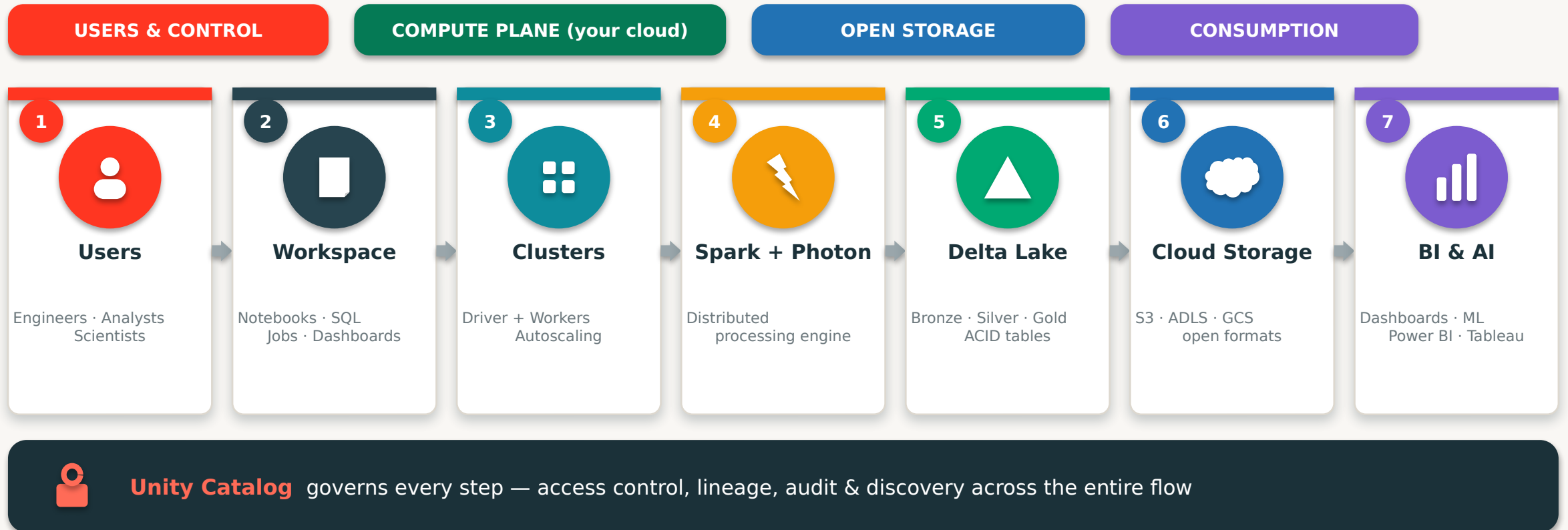
Search, tags & Delta Sharing — open data sharing across organizations.



Protection & Compliance

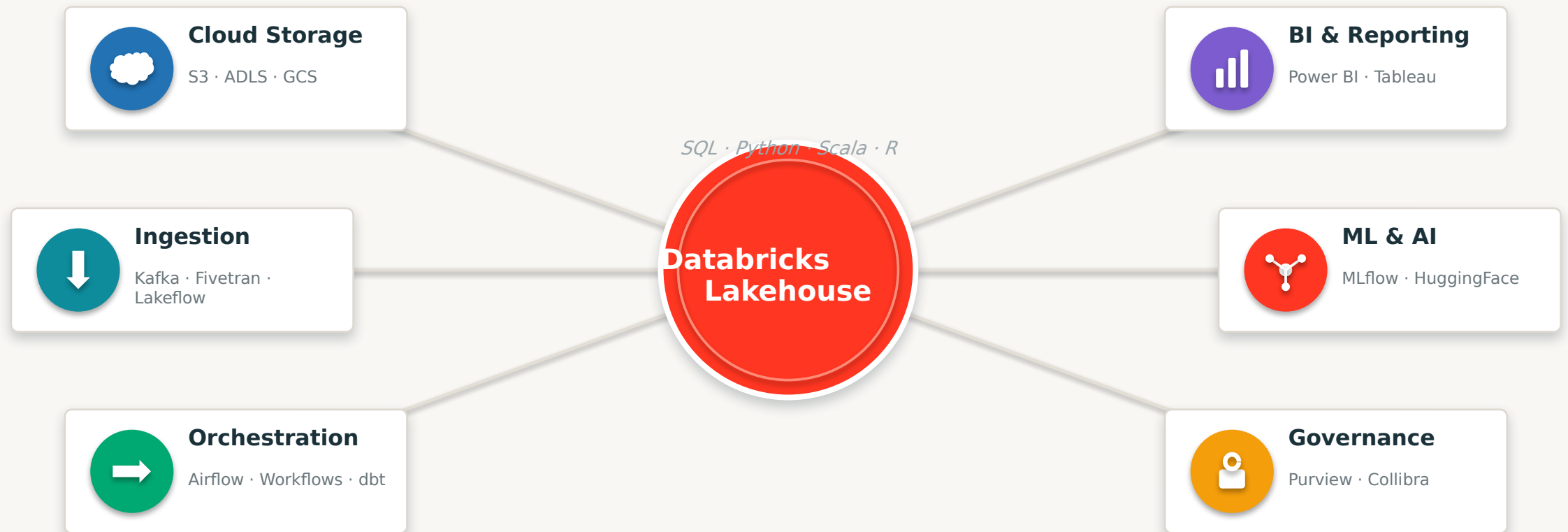
Encryption, network isolation, GDPR / HIPAA / SOC 2 readiness.

End-to-End Data Flow



From a user's query to a governed dashboard — one platform, your data, end to end.

Integration Architecture — An Open Hub



Key Takeaways

1 Big Data Fundamentals

- Scale OUT, not up — distributed compute
- The 5 V's define the problem space
- Sources → ... → Visualization pipeline

2 Databricks

- Unified Lakehouse for data + AI
- Warehouse reliability + lake openness
- Spark, Delta, MLflow, Unity Catalog

3 Architecture

- Control plane vs. your compute plane
- Delta + Medallion = trusted data
- Unity Catalog governs end to end



The Lakehouse unifies all data, analytics & AI on one open, governed platform.



Your data stays in your cloud — Databricks manages the experience, you keep control.



Delta Lake + the medallion pattern turn raw files into trusted tables.



One governance model (Unity Catalog) secures every asset, every step.

Knowledge Check & Interview Questions



KNOWLEDGE CHECK

- Q1. Name the 5 V's of Big Data — which two decide success?
- Q2. Scale UP vs scale OUT — which is Big Data, and why?
- Q3. What does Delta Lake add on top of Parquet files?
- Q4. Control plane vs compute plane — where does your data live?
- Q5. List the three medallion layers in order.



COMMON INTERVIEW QUESTIONS

- Explain the Lakehouse — how does it differ from a warehouse and a lake?
- How does Spark execute a job? (driver, executors, partitions)
- What problems does Unity Catalog solve?
- Describe ACID & time travel in Delta Lake.
- When would you use job vs all-purpose vs serverless clusters?

Thank You

You now have the full picture — from Big Data fundamentals to the Databricks Lakehouse architecture.

KEEP LEARNING

docs.databricks.com

[Databricks Academy](#)

[Free Community Edition](#)

[Delta Lake & Spark docs](#)



Big Data



Lakehouse



Spark



Delta



Medallion



Unity Catalog



Questions & discussion welcome